Resonant Order Jets and Convergent Inverse Transseries

for Polylogarithms and Logarithmically Weighted Tails

Research draft prepared for Vladimir Reshetnikov
A contribution for the ProveIt research collection

12 October 2026

Abstract

We study the inverse of a polylogarithmic endpoint deficit and the changes in its transseries when the order passes through an integer. An exact pole-cancellation formula expresses every resonant order derivative in terms of complete Bell polynomials, harmonic numbers, zeta values and Stieltjes constants. For higher integer orders, the inverse is a convergent logarithmic series with a sharp degree bound. Its highest logarithmic coefficients obey a universal Fuss–Catalan law, including arbitrary fixed nonnegative integer logarithmic weights.

At the critical order two, we construct a single holomorphic inverse chart for the weighted deficit $\sum_{\nu\geq 1}(\log\nu)^k(1-e^{-\nu t})/\nu^{2+\varepsilon}$. It is uniform when $\varepsilon\log(1/y)$ stays in a compact real interval and converges in three independent small variables. The construction gives all inverse-logarithmic and exponential blocks, an explicit first correction, and the first possible exponentially smaller block.

We also give a self-contained global contour argument for coefficients of the ordinary compositional inverse. It identifies two missing normalization factors in the author-hosted 10 February 2025 note of Krattenthaler and Rivoal. For orders at least three, the entire coefficient expansion has a sharp logarithmic support boundary with explicit Fuss–Catalan coefficients. For the dilogarithm, an all-orders expansion in an exact Lambert variable has coefficients given by derivatives of the reciprocal Gamma function at negative integers. Convergent local charts are carefully distinguished from these outer asymptotic expansions. The accompanying code provides exact algebraic checks and independent high-precision numerical diagnostics.

Status. The results are proved below in conventional mathematical prose. The proposed contributions are measured against the explicitly inspected repository material and primary sources; worldwide priority is not claimed. No Lean formalization is claimed. The numerical diagnostics are not interval certificates.

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1 Research question, contributions, and provenance

The natural common ground between polylogarithms and transseries is not only the large-argument expansion of a special function. It is also the local inverse problem at a singular endpoint. The classical formula

$$\mathrm{Li}_s(e^{-t}) = \Gamma(1-s)t^{s-1} + \sum_{j\geq 0} \zeta(s-j)\frac{(-t)^j}{j!}$$

contains a fractional power for nonintegral s. At a positive integer, the Gamma pole and one zeta pole cancel and leave a logarithm. Inversion turns this elementary collision into a structured family of power–logarithm series. Differentiation in s raises the logarithmic degree, while the limit $s \to 2$ changes the leading inverse scale.

Three distinctions guide the paper. First, an exact identity of holomorphic germs is stronger than a formal reversion. Second, a convergent expansion in several independent small variables may become a transseries after those variables are related. Third, coefficient extraction from such a germ generally produces a new asymptotic expansion; convergence of the original chart does not establish convergence of that outer expansion.

1.1 Main results

For orientation, the principal conclusions are the following.

1. **All resonant order jets.** Theorem 3.1 gives an exact, normally convergent formula at every positive integer order. Its logarithmic polynomial is an explicit complete Bell polynomial. This is a systematic derivation from the classical Jonquière expansion, with all signs and removable limits fixed.

2. **Weighted logarithmic inverse and its sharp boundary.** For $m \geq 2$, $k \geq 0$, and

$$D_{m,k}(t) = \sum_{\nu\geq 1} \frac{(\log \nu)^k(1-e^{-\nu t})}{\nu^{m+1}},$$

   Theorem 4.1 proves a convergent expansion $t(y) = \sum_{N\geq 1} y^N T_N(\log y)$, with

$$\deg T_N \leq (k+1)\left\lfloor \frac{N-1}{m-1} \right\rfloor.$$

   The bound is attained on an infinite set of indices. The corresponding coefficients are given exactly by Fuss–Catalan numbers.

3. **Uniform weighted transition through order two.** Theorem 6.1 gives a convergent analytic chart for $D_{\varepsilon,k}(t) = y$, uniformly in $\tau = \varepsilon \log(1/y)$ on compact real intervals. The leading scale is

$$t \sim \frac{y}{T^{k+1}H_k(\tau)}, \qquad T = \log(1/y), \qquad H_k(\tau) = \int_0^1 u^k e^{-\tau u}\, du.$$

   All three small variables and their exact implicit equation are given. The first correction and the first exponential block are explicit.

4. **Corrected and refined inverse coefficient asymptotics.** If $\mathrm{Li}_s(\ell_s(w)) = w$ near zero and $a_n^{(s)} = [w^n]\ell_s(w)$, then

$$a_n^{(2)} \sim -\frac{\zeta(2)^{1-n}}{n^2(\log n)^2}, \qquad a_n^{(s)} \sim -\frac{\zeta(s)^{s-1-n}}{\zeta(s-1)^s n^s} \quad (s \in \{3,4,\ldots\}).$$

Theorem 9.1 gives an all-orders refinement and its highest-logarithm coefficients. Theorem 10.1 gives the dilogarithmic expansion in a Lambert variable, with a reciprocal Gamma coefficient formula.

1.2 Relation to the inspected ProveIt material

The repository was read at commit

`9ec0f0406afe7a5475b03ff65ed91f7e283979b8`.

The scope of the inspection is recorded in the accompanying provenance file. It includes the Polylogarithms manuscript chapters and the relevant Transseries reports, rather than a claim to have audited every proof in the two collections.

The Polylogarithms manuscript already derives the $s = 1$ first order jet in its Tornheim evaluation chapter. It also uses Gamma–harmonic generators and finite-part resonance calculations in its periodic collision and Stieltjes chapters [6]. These are direct antecedents of Section 3.

The Transseries reports already establish several convergent Lambert charts, explicit Gamma–zeta pole cancellation near order three, and analytic inverse transport [7, 8, 9]. The report on slowly varying action tails treats fixed $1 < \alpha < 2$ and the exact tail

$$a(-\partial_\alpha)^k \operatorname{Li}_{1+\alpha}(z) + P(z).$$

Its Research Question 5 leaves joint endpoint and logarithmic-weight limits open [10]. Our critical theorem addresses the *fixed integer weight, bare inverse* part at the lower endpoint. It does not settle the associated nonlinear feedback problem, its coefficient limits, growing weights, or cutoff laws.

The precise weighted inverse support laws, the uniform order-two chart in the form used here, and the all-orders inverse coefficient refinements are the proposed additions. The implicit-function theorem, Lagrange inversion, the Jonquière expansion, and singularity transfer are classical tools, not new claims.

1.3 The external normalization issue

Krattenthaler and Rivoal's author-hosted note *On the compositional inverse of polylogarithms*, dated 10 February 2025, studies the same ordinary inverse coefficients [4]. In the inspected version, both leading constants in Theorem 1 lose factors during the calculation. Section 11 locates the losses precisely. We establish the corrected constants independently, including the required global contour control. The correction is limited to that identified version; it is not a claim about an uninspected later revision.

2 Conventions and the classical local expansion

For $|z| < 1$ and $s \in \mathbb{C}$, put

$$\operatorname{Li}_s(z) = \sum_{\nu \geq 1} \frac{z^\nu}{\nu^s}.$$

The principal z-branch is defined on $\Lambda = \mathbb{C} \setminus [1, \infty)$. Near $t = 0$, we use $\operatorname{Li}_s(e^{-t})$ continued from $t > 0$, with $-\pi < \arg t < \pi$ and $t^\alpha = e^{\alpha \operatorname{Log} t}$. Closed subsector statements always mean $|\arg t| \leq \pi - \theta$, where $\theta > 0$ is fixed. A small neighborhood can be reduced whenever necessary to retain the stated branch.

We write $H_m = \sum_{j=1}^{m} 1/j$, $H_m^{(r)} = \sum_{j=1}^{m} j^{-r}$, and $H_0^{(r)} = 0$. The Stieltjes constants have the convention

$$\zeta(1+\varepsilon) = \frac{1}{\varepsilon} + \sum_{r\geq 0} \frac{(-1)^r \gamma_r}{r!} \varepsilon^r, \qquad \gamma_0 = \gamma. \tag{2.1}$$

The symbol $(\log 1)^0$ is interpreted as 1 when it occurs in an unweighted series. For $k \geq 1$, $(\log 1)^k = 0$. All order derivatives are taken with the argument held fixed.

Proposition 2.1 (Classical Jonquière expansion)**.** *For $s \notin \{1, 2, \ldots\}$ and $0 < |t| < 2\pi$ on the specified branch,*

$$\mathrm{Li}_s(e^{-t}) = \Gamma(1-s)t^{s-1} + \sum_{j=0}^{\infty} \zeta(s-j)\frac{(-t)^j}{j!}. \tag{2.2}$$

Away from the exceptional orders, the power series converges normally on compact subsets of $|t| < 2\pi$ and locally uniformly in s.

This is the classical identity recorded in DLMF 25.12.12 [1]. It is not merely an asymptotic series. At a positive integer order the two meromorphic summands that collide must be combined before the parameter is specialized or differentiated.

Normal-convergence detail used below. For s in a compact set and j large, the zeta functional equation expresses $\zeta(s-j)$ as a product of $2^{s-j}\pi^{s-j-1}$, a bounded trigonometric factor, $\Gamma(1-s+j)$, and $\zeta(1-s+j)$. After division by $j!$, the Gamma quotient grows at most polynomially in j, uniformly on that compact set. Thus the coefficients are bounded by $C(2\pi)^{-j}(1+j)^M$. The same estimate with a different polynomial, or Cauchy's formula in the order parameter, controls every fixed order derivative. Removing the single colliding index leaves a normally convergent, jointly holomorphic remainder near any fixed positive integer. □

Definition 2.2 (Convergent transseries chart)**.** In this paper a convergent chart is an identity

$$f(y) = M(y)\, V(g_1(y), \ldots, g_r(y)),$$

where V is holomorphic near the origin and each $g_j(y) \to 0$ in the stated domain. Its ordinary multivariable Taylor series converges absolutely after substitution. When the g_j are powers, logarithmic expressions, or exponentials, we call the resulting representation a convergent transseries.

This definition provides a concrete analytic realization and a coefficient algorithm. It does not assert that a later expansion obtained by saddle integration or coefficient extraction converges.

3 Exact resonance cancellation and every order jet

Fix $m \in \mathbb{N}_0$, so that the resonant order is $s = m+1$. Define

$$G_m(\varepsilon) = \frac{\Gamma(1-\varepsilon)}{\prod_{j=1}^{m}(1+\varepsilon/j)}, \qquad G_m(0) = 1, \tag{3.1}$$

where an empty product equals one. The Gamma recurrence gives

$$\Gamma(-m-\varepsilon) = \frac{(-1)^{m+1}}{m!\varepsilon}\, G_m(\varepsilon).$$

If $L = \operatorname{Log} t$, the resonant pair in (2.2) is therefore exactly

$$\Gamma(-m-\varepsilon)t^{m+\varepsilon} + \zeta(1+\varepsilon)\frac{(-t)^m}{m!} = \frac{(-t)^m}{m!}\left\{Z(\varepsilon) - \frac{G_m(\varepsilon)e^{\varepsilon L}-1}{\varepsilon}\right\}, \tag{3.2}$$

where $Z(\varepsilon) = \zeta(1+\varepsilon) - 1/\varepsilon$. Both apparent singularities on the right are removable.

Let $\mathcal{B}_r(x_1,\dots,x_r)$ be the complete exponential Bell polynomial, normalized by

$$\exp\left(\sum_{j\ge1} x_j \frac{u^j}{j!}\right) = \sum_{r\ge0} \mathcal{B}_r(x_1,\dots,x_r)\frac{u^r}{r!}. \tag{3.3}$$

For $r \ge 0$, put

$$\begin{aligned} X_1 &= L + \gamma - H_m, \\ X_j &= (j-1)!\left\{\zeta(j) + (-1)^j H_m^{(j)}\right\}, \qquad 2 \le j \le r+1. \end{aligned} \tag{3.4}$$

Theorem 3.1 (All resonant order derivatives)**.** *For $m, r \in \mathbb{N}_0$, on the stated slit disc,*

$$\begin{aligned} \partial_s^r \operatorname{Li}_s(e^{-t})\Big|_{s=m+1} = &\sum_{\substack{j\ge0\\ j\ne m}} \zeta^{(r)}(m+1-j)\frac{(-t)^j}{j!} \\ &+ \frac{(-t)^m}{m!}\left\{(-1)^r\gamma_r - \frac{\mathcal{B}_{r+1}(X_1,\dots,X_{r+1})}{r+1}\right\}. \end{aligned} \tag{3.5}$$

The power series converges normally for $|t| < 2\pi$. The displayed logarithmic polynomial has degree exactly $r+1$.

Proof. The logarithm of G_m is convergent near $\varepsilon = 0$:

$$\log G_m(\varepsilon) = (\gamma - H_m)\varepsilon + \sum_{j\ge2} \frac{\zeta(j) + (-1)^j H_m^{(j)}}{j}\varepsilon^j. \tag{3.6}$$

This follows by subtracting the finite sum $\sum_{j=1}^m \log(1+\varepsilon/j)$ from the standard expansion of $\log\Gamma(1-\varepsilon)$. Equations (3.3) and (3.6) give

$$G_m(\varepsilon)e^{\varepsilon L} = \sum_{q\ge0} \mathcal{B}_q(X_1,\dots,X_q)\frac{\varepsilon^q}{q!}.$$

The r-th derivative at zero of $(G_m(\varepsilon)e^{\varepsilon L}-1)/\varepsilon$ is $\mathcal{B}_{r+1}/(r+1)$. The corresponding derivative of Z is $(-1)^r\gamma_r$. Substitute in (3.2) and differentiate the normally convergent remainder using Proposition 2.1. The leading term of $\mathcal{B}_{r+1}$ is X_1^{r+1}, proving the degree assertion. □

Example 3.2 (Two checks on the normalization)**.** For $r = 0$, the theorem reduces to

$$\operatorname{Li}_{m+1}(e^{-t}) = \sum_{\substack{j\ge0\\ j\ne m}} \zeta(m+1-j)\frac{(-t)^j}{j!} + \frac{(-t)^m}{m!}(H_m - \operatorname{Log} t). \tag{3.7}$$

For $m = 0, r = 1$, it gives

$$\begin{aligned} \partial_s \operatorname{Li}_s(e^{-t})\Big|_{s=1} = &-\gamma_1 - \frac{\gamma^2+\zeta(2)}{2} - \gamma\operatorname{Log} t - \frac{(\operatorname{Log} t)^2}{2} \\ &+ \sum_{j\ge1} \zeta'(1-j)\frac{(-t)^j}{j!}. \end{aligned}$$

This agrees with the accepted Tornheim calculation in the repository [6]; it is an internal normalization check, not an additional claim of novelty.

Remark 3.3 (Uniformity before expansion). Formula (3.2) remains exact when εL is of order one. A finite Taylor polynomial in ε generally does not approximate it uniformly in that regime, because successive derivatives contribute successive powers of L. Keeping the entire divided exponential is the correct starting point for a uniform transition. Section 6 does this before inverting.

3.1 A precise monodromy observation

In (3.5), continuation once around $t = 0$ replaces L by $L + 2\pi i$, while the analytic power series stays unchanged. Consequently the monodromy of the r-th order jet is t^m times an explicitly known polynomial of degree r. For the unweighted polylogarithm, the change is $-2\pi i(-t)^m/m!$. This is ordinary local monodromy. The convergent expansions here do not acquire a Borel summation ambiguity, and calling this calculation a Stokes jump would conflate two different notions.

4 Convergent inverses at higher integer orders

For $m \geq 2$ and $k \geq 0$, define

$$D_{m,k}(t) = \sum_{\nu \geq 1} \frac{(\log \nu)^k}{\nu^{m+1}}(1 - e^{-\nu t}). \tag{4.1}$$

The series and its first t-derivative converge at $t = 0+$. It is strictly increasing on the positive axis, and

$$D'_{m,k}(0+) = a := (-1)^k \zeta^{(k)}(m) > 0. \tag{4.2}$$

Theorem 3.1 supplies the exact decomposition

$$D_{m,k}(t) = tA(t) + t^m P(\operatorname{Log} t), \tag{4.3}$$

where A is holomorphic for $|t| < 2\pi$, $A(0) = a$, and

$$A(t) = \sum_{\substack{j \geq 1 \\ j \neq m}} \frac{(-1)^{k+j+1}\zeta^{(k)}(m+1-j)}{j!}\, t^{j-1}, \tag{4.4}$$

$$P(L) = \frac{(-1)^{m+k}}{m!}\left\{ \frac{\mathcal{B}_{k+1}(X_1, \ldots, X_{k+1})}{k+1} - (-1)^k \gamma_k \right\}. \tag{4.5}$$

Here X_j use m as in (3.4). In particular, with $d = k + 1$,

$$\deg P = d, \qquad b := [L^d]P(L) = \frac{(-1)^{m+k}}{m!(k+1)} \neq 0. \tag{4.6}$$

Theorem 4.1 (A convergent inverse and its sharp logarithmic boundary)**.** *More generally, suppose*

$$D(t) = tA(t) + t^m P(\operatorname{Log} t), \qquad m \geq 2,$$

where A is holomorphic near zero with real Taylor coefficients, $A(0) = a > 0$, and P is a real polynomial of degree $d \geq 1$ with leading coefficient $b \neq 0$. Put $r = m - 1$. The inverse with $t(y) \sim y/a$ has the following properties.

(i) There is a holomorphic function V near the origin of $\mathbb{C}^{d+2}$, with $V(0) = 1$, such that

$$t(y) = \frac{y}{a} V\big(y, y^r, y^r \operatorname{Log} y, \dots, y^r (\operatorname{Log} y)^d\big). \tag{4.7}$$

The identity holds for small positive y, and on sufficiently small closed subsectors of the slit y-plane.

(ii) There is a canonically regrouped, absolutely convergent expansion

$$t(y) = \sum_{N \geq 1} y^N T_N(\operatorname{Log} y), \qquad \deg T_N \leq d \left\lfloor \frac{N-1}{r} \right\rfloor. \tag{4.8}$$

For every fixed $N \geq 1$, its remainder is

$$O\left(|y|^{N+1} (1 + |\operatorname{Log} y|)^{d\lfloor N/r \rfloor}\right). \tag{4.9}$$

(iii) For $q \in \mathbb{N}_0$, the highest-logarithm corner is

$$[y^{1+rq} (\operatorname{Log} y)^{dq}]\, t(y) = \frac{(-b)^q}{a^{mq+1}} \frac{1}{rq+1} \binom{mq}{q}. \tag{4.10}$$

In particular, the degree bound in (4.8) is attained at every $N = 1 + rq$.

Proof. Choose $\log v$ near $v = 1$, and define

$$Q_j(v) = [u^j] P(u - \log a + \log v), \qquad 0 \leq j \leq d.$$

These functions are holomorphic, and $Q_d(v) = b$ is constant. Introduce independent variables $Y, Z_0, \dots, Z_d$ and the equation

$$F(v; Y, Z) = \frac{v}{a} A(Y v/a) + a^{-m} v^m \sum_{j=0}^{d} Z_j Q_j(v) - 1 = 0. \tag{4.11}$$

At $(v; Y, Z) = (1; 0, 0)$, $F = 0$ and $F_v = 1$. The holomorphic implicit-function theorem produces a unique local solution $v = V(Y, Z)$. On substituting $Y = y$, $Z_j = y^r (\operatorname{Log} y)^j$, and $t = yv/a$, equation (4.11) becomes $D(t) = y$. For positive small y the solution is positive and is the inverse selected by $t \sim y/a$. On a closed subsector, the logarithm identity $\operatorname{Log} t = \operatorname{Log} y - \log a + \log v$ remains valid after shrinking the radius.

Expand V in its ordinary Taylor series:

$$V(Y, Z) = \sum_{p \geq 0,\, \beta \in \mathbb{N}_0^{d+1}} v_{p,\beta} Y^p Z^\beta.$$

After substitution, a term in t has power and logarithmic degree

$$N = 1 + p + r|\beta|, \qquad J = \sum_{j=0}^{d} j\beta_j \leq d|\beta|. \tag{4.12}$$

For each N, only finitely many such multiindices occur. Absolute convergence permits their regrouping, and (4.12) proves the degree bound.

For clarity, a useful quantitative version follows directly from Cauchy estimates: for some $C, K > 0$,

$$|T_N(L)| \leq C K^N (1 + |L|)^{d\lfloor (N-1)/r \rfloor}.$$

The finite number of relevant multiindices is polynomial in N; increasing K absorbs it. Thus, whenever $K|y|(1+|L|)^d < 1$,

$$\left| t(y) - \sum_{n=1}^{N} y^n T_n(L) \right| \le \frac{CK^{N+1}|y|^{N+1}(1+|L|)^{d\lfloor N/r \rfloor}}{1-K|y|(1+|L|)^d}. \tag{4.13}$$

This proves the stated remainder. It also proves uniqueness of the regrouped polynomials: at the first different N, their nonzero polynomial difference in $\operatorname{Log} y$ could not be bounded by y times any fixed power of $|\operatorname{Log} y|$.

Finally, at $N = 1 + rq$ and $J = dq$, both inequalities in (4.12) must be equalities. Hence $p = 0$, $|\beta| = q$, and the only nonzero component is $\beta_d = q$. Set $Y = Z_0 = \cdots = Z_{d-1} = 0$, $Z_d = z$. The relevant slice $W(z) = V(0, 0, \ldots, z)$ satisfies

$$W + \frac{b}{a^m} z W^m = 1.$$

Writing $U = 1 - W$, we have $U = (b/a^m)z(1-U)^m$. Ordinary Lagrange inversion gives

$$[z^q]W(z) = \left(-\frac{b}{a^m}\right)^q \frac{1}{(m-1)q+1} \binom{mq}{q}.$$

The outside factor y/a proves (4.10). □

Remark 4.2 (Why the auxiliary redundancy is harmless). On the physical curve, $Z_0 = Y^r$, so the independent lift is not a minimal coordinate system. Its uniquely defined implicit equation is convenient for coefficient extraction. The regrouped polynomials T_N, which represent the actual inverse, are intrinsic by the uniqueness argument. No uniqueness claim is made for every possible ambient analytic lift.

4.1 Examples and an algebraic slice

For the ordinary trilogarithm, $m = 2, k = 0$, $a = \zeta(2)$, $b = 1/2$. The first highest-logarithm coefficients of t are

$$\frac{1}{a}, \quad -\frac{1}{2a^3}, \quad \frac{1}{2a^5}, \quad -\frac{5}{8a^7},$$

at the respective monomials $y, y^2 \log y, y^3(\log y)^2, y^4(\log y)^3$. For the logarithmically weighted trilogarithmic deficit $m = 2, k = 1$,

$$a = -\zeta'(2), \qquad b = -\frac{1}{4},$$

the corresponding top monomials are $y^{q+1}(\log y)^{2q}$, with positive coefficients

$$\frac{1}{4^q a^{2q+1}} \frac{1}{q+1} \binom{2q}{q}.$$

Every lower coefficient of A and P may change without changing this boundary law, provided a, b, m, d stay fixed.

The isolated slice $W + czW^m = 1$, $c = b/a^m$, has its first algebraic critical point at

$$W_* = \frac{m}{m-1}, \qquad z_* = -\frac{(m-1)^{m-1}}{m^m c}.$$

These values explain the Fuss–Catalan growth of the corner sequence. They are singularity data for that *slice*. They do not determine the full inverse's global singularities after the dependent coordinates in (4.7) are restored.

5 The neighboring nonintegral inverse regimes

It is useful to contrast the resonant logarithmic chart with the nonintegral cases without hiding their dependence on the order. Write

$$\Delta_s(t) = \zeta(s) - \mathrm{Li}_s(e^{-t}), \qquad s > 1.$$

For nonintegral s, let $\alpha = s - 1$ and $b = -\Gamma(-\alpha)$. Then

$$\Delta_s(t) = bt^\alpha + tA_s(t), \qquad A_s(0) = \zeta(s-1), \tag{5.1}$$

with an analytic A_s.

Proposition 5.1 (Two finite-generator nonintegral charts)**.** *For a fixed real nonintegral $s > 1$, the positive inverse has the following convergent representations.*

(i) If $s > 2$, put $a = \zeta(s-1) > 0$. Then

$$t(y) = \frac{y}{a} V_s(y, y^{s-2}), \qquad V_s(0,0) = 1,$$

where V_s is holomorphic near the origin. The support is contained in $\{y^{1+p+(s-2)q} : p, q \in \mathbb{N}_0\}$.

(ii) If $1 < s < 2$, put $x = (y/b)^{1/\alpha}$, so that $b > 0$. Then

$$t(y) = x\, W_s(x, x^{1-\alpha}), \qquad W_s(0,0) = 1,$$

with W_s holomorphic near the origin.

Compatible local complex branches satisfy the same identities.

Proof. For (i), put $t = yv/a$ in (5.1). After division by y, the equation is holomorphic in $(y, y^{\alpha-1}, v)$ near $(0,0,1)$, and its derivative in v is one. For (ii), put $t = xv$ and divide by $bx^\alpha = y$. The equation is

$$1 = v^\alpha + \frac{x^{1-\alpha}}{b}\, vA_s(xv),$$

whose v-derivative at the base point is $\alpha \neq 0$. The implicit-function theorem proves both claims. □

If the exponents are rational, these representations can be collected into convergent Puiseux series. For irrational exponents they remain convergent generalized power series with finitely generated support. At higher positive integers, the nonzero logarithmic corner in Theorem 4.1 obstructs a convergent Puiseux representation. At order two, even the leading term changes from a constant times y or a fractional power to $y/\log(1/y)$. None of the fixed-s charts in this section is asserted to be uniform as $s \to 2$; the next section supplies the missing uniform chart.

6 A uniform weighted inverse through order two

Fix $k \in \mathbb{N}_0$. For $\varepsilon > -1$, $t > 0$, define

$$D_{\varepsilon,k}(t) = \sum_{\nu \geq 1} \frac{(\log \nu)^k (1 - e^{-\nu t})}{\nu^{2+\varepsilon}} = (-\partial_\varepsilon)^k \Delta_{2+\varepsilon}(t). \tag{6.1}$$

This notation differs from $D_{m,k}$ in Section 4: here the first subscript is the displacement from order two. The series converges and is strictly increasing in t. Its limit as $t \to \infty$ is $(-1)^k \zeta^{(k)}(2+\varepsilon) > 0$. Thus for small positive y there is a unique positive inverse.

Define entire functions

$$H_k(\tau) = \int_0^1 u^k e^{-\tau u}\, du, \tag{6.2}$$

$$J_k(\tau, U) = \int_0^U u^k e^{-\tau u}\, du. \tag{6.3}$$

The second integral is path independent. In particular,

$$H_k(\tau) = \frac{k!}{\tau^{k+1}} \left(1 - e^{-\tau} \sum_{j=0}^{k} \frac{\tau^j}{j!}\right), \qquad H_k(0) = \frac{1}{k+1}. \tag{6.4}$$

For real τ, $H_k(\tau) > 0$.

Theorem 6.1 (Convergent uniform critical chart)**.** *For every compact real interval K, there are a complex neighborhood Ω of K, $\delta > 0$, and a holomorphic function*

$$V_k(\tau; h, q, p), \qquad \tau \in \Omega, \quad |h|, |q|, |p| < \delta,$$

such that $V_k(\tau; 0, 0, 0) = H_k(\tau)^{-1}$. Let

$$T = \log(1/y), \qquad \tau = \varepsilon T.$$

Uniformly for $\tau \in K$ and all sufficiently large T, the exact positive solution of $D_{\tau/T,k}(t) = y$ is

$$\boxed{t = \frac{y}{T^{k+1}} V_k\left(\tau; \frac{1}{T}, \frac{\log T}{T}, \frac{y}{T^{k+1}}\right).} \tag{6.5}$$

The Taylor series in h, q, p converges absolutely, uniformly on smaller polydiscs. In particular the leading equivalence

$$t \sim \frac{y}{T^{k+1} H_k(\tau)} \tag{6.6}$$

is uniform on K.

6.1 An exact equation without colliding poles

For the proof, define the holomorphic germs

$$Z(\varepsilon) = \zeta(1+\varepsilon) - \frac{1}{\varepsilon}, \qquad G(\varepsilon) = \Gamma(-1-\varepsilon) - \frac{1}{\varepsilon}. \tag{6.7}$$

Then $Z(0) = \gamma$, $G(0) = \gamma - 1$. The letter G here denotes the *regular part*; it is not the normalized G_m from (3.1). Set

$$R(t, \varepsilon) = \sum_{j \geq 2} \frac{(-1)^{j+1} \zeta(2+\varepsilon-j)}{j!}\, t^{j-1}, \qquad R_k = (-\partial_\varepsilon)^k R. \tag{6.8}$$

For each $\eta < 1$, R is jointly holomorphic on $|t| < 2\pi$, $|\varepsilon| < \eta$, and $R(0, \varepsilon) = 0$. With $L = -\operatorname{Log} t$, exact cancellation gives

$$\frac{D_{\varepsilon,0}(t)}{t} = \frac{1 - e^{-\varepsilon L}}{\varepsilon} + Z(\varepsilon) - G(\varepsilon) e^{-\varepsilon L} + R(t, \varepsilon).$$

Because the divided exponential is $\int_0^L e^{-\varepsilon u}\,du$, differentiation at fixed t gives the exact identity

$$\frac{D_{\varepsilon,k}(t)}{t} = L^{k+1}H_k(\varepsilon L) + (-1)^k Z^{(k)}(\varepsilon) - e^{-\varepsilon L}\sum_{j=0}^{k}\binom{k}{j}(-1)^j G^{(j)}(\varepsilon)L^{k-j} + R_k(t,\varepsilon). \tag{6.9}$$

At $\varepsilon = 0$, this is a finite polynomial in L, plus a convergent analytic remainder. For example,

$$\frac{D_{0,1}(t)}{t} = \frac{L^2}{2} - (\gamma-1)L + 1 - \gamma + \frac{\gamma^2+\zeta(2)}{2} + \gamma_1 + R_1(t,0). \tag{6.10}$$

Proof of Theorem 6.1. Write $t = pv$, and introduce

$$U = 1 + (k+1)q - h\log v.$$

Under the actual substitutions in (6.5), $L = TU$, $\varepsilon = \tau h$, and $\varepsilon L = \tau U$. Define

$$B_k(\tau;h,q,p,v) = J_k(\tau,U) + (-1)^k h^{k+1}Z^{(k)}(\tau h) - e^{-\tau U}\sum_{j=0}^{k}\binom{k}{j}(-1)^j h^{j+1}G^{(j)}(\tau h)U^{k-j} + h^{k+1}R_k(pv,\tau h), \tag{6.11}$$

$$\Phi_k(\tau;h,q,p,v) = vB_k(\tau;h,q,p,v) - 1. \tag{6.12}$$

Since $J_k(\tau,U) = U^{k+1}H_k(\tau U)$, equation (6.9) says exactly that $D_{\varepsilon,k}(t) = y$ is equivalent to $\Phi_k = 0$.

For real τ, choose $\log v$ near the positive base value $v_0(\tau) = 1/H_k(\tau)$. At $h = q = p = 0$,

$$\Phi_k = vH_k(\tau) - 1, \qquad \partial_v\Phi_k = H_k(\tau) \neq 0.$$

The holomorphic implicit-function theorem gives V_k locally in τ and the three small variables. Positivity of H_k, compactness of K, and uniqueness on overlaps give a common small polydisc over a thin complex neighborhood of K. For sufficiently large T, all physical coordinates lie in it. The solution is real and positive by continuity, and strict monotonicity of $D_{\varepsilon,k}$ identifies it with the positive inverse. Absolute and locally uniform Taylor convergence is ordinary holomorphic power-series convergence. □

Remark 6.2 (The complex parameter obstruction). The same construction is valid locally at any complex τ for which $H_k(\tau) \neq 0$. A uniform complex compact set requires a compatible holomorphic logarithm of $1/H_k$ on a neighborhood. At a zero of H_k, the limiting equation $vH_k(\tau) = 1$ has no finite solution, so this normalization must be replaced. For $k = 0$, the nonzero exceptional points are $2\pi i\mathbb{Z}\setminus\{0\}$. For $k > 0$, they satisfy $e^\tau = \sum_{j=0}^{k}\tau^j/j!$, excluding the removable zero at zero. We make no claim that the same scale works at these exceptional points.

6.2 First correction and a complete coefficient recursion

Put

$$g_k(\tau) = e^{-\tau}\{\log H_k(\tau) - \gamma + 1\} + \mathbf{1}_{\{k=0\}}\gamma. \tag{6.13}$$

Differentiating (6.12) at the base point gives

$$[q]V_k = -\frac{(k+1)e^{-\tau}}{H_k(\tau)^2}, \qquad [h]V_k = -\frac{g_k(\tau)}{H_k(\tau)^2}, \qquad [p]V_k = 0. \tag{6.14}$$

Therefore

Corollary 6.3 (An explicit uniform first correction)**.** *Uniformly for real τ in a fixed compact interval,*

$$t = \frac{y}{T^{k+1}H_k(\tau)}\left[1 - \frac{(k+1)e^{-\tau}\log T + g_k(\tau)}{H_k(\tau)T} + O_K\left(\frac{(\log T)^2}{T^2}\right)\right]. \tag{6.15}$$

The implied constant can depend on the fixed integer k.

Proof. In (6.11), $J_{k,U}(\tau,1) = e^{-\tau}$, and $U - 1 = (k+1)q - h\log v$. At the base value $\log v_0 = -\log H_k$, this yields the first two coefficients in (6.14); the regular Z term is first order in h only for $k = 0$. The function R_k vanishes at $t = 0$, so there is no pure p term. Taylor's theorem and $h, q, p \to 0$ give the relative remainder. □

At $\tau = 0$, the formula becomes

$$t = \frac{(k+1)y}{T^{k+1}}\left[1 - \frac{k+1}{T}\left\{(k+1)\log T + 1 - \gamma - \log(k+1) + \mathbf{1}_{\{k=0\}}\gamma\right\} + O\left(\frac{(\log T)^2}{T^2}\right)\right]. \tag{6.16}$$

For $k = 0$, this reads

$$t = \frac{y}{T}\left[1 - \frac{\log T + 1}{T} + O\left(\frac{(\log T)^2}{T^2}\right)\right].$$

Every further coefficient is finite and algorithmic. Let $x = (h, q, p)$ and write $V_k = \sum_{\nu\in\mathbb{N}_0^3} c_\nu(\tau)x^\nu$, with $c_0 = H_k^{-1}$. If the coefficients below total degree N are known and $V_{<N}$ is their sum, then

$$c_\nu(\tau) = -\frac{1}{H_k(\tau)}[x^\nu]\,\Phi_k(\tau; x, V_{<N}), \qquad |\nu| = N. \tag{6.17}$$

Only the linear term $H_k c_\nu$ can involve a new coefficient at that degree. This proves the recursion, rather than merely suggesting a formal iterative procedure.

After substitution, the allowed monomials are

$$\frac{y^{c+1}(\log T)^b}{T^{(k+1)(c+1)+a+b}}, \qquad a, b, c \in \mathbb{N}_0. \tag{6.18}$$

The coefficients are holomorphic functions of τ, built from the functions in (6.11). Stieltjes constants and Gamma/zeta derivatives occur naturally in this coefficient field.

6.3 The first exponential block

Define the complete logarithmic block

$$t_{\log}(y,\tau) = \frac{y}{T^{k+1}}V_k\left(\tau; T^{-1}, T^{-1}\log T, 0\right).$$

This expression retains *all* terms of the convergent $p = 0$ chart. The p-dependence in (6.11) enters only through $h^{k+1}R_k(pv, \tau h)$, with $R_k(0,\varepsilon) = 0$. Analytic implicit differentiation consequently shows that $V_k(h,q,p) - V_k(h,q,0)$ is divisible by $h^{k+1}p$. Since

$$\partial_t R_k(0,0) = \frac{(-1)^{k+1}}{2}\zeta^{(k)}(0),$$

we obtain the exact support statement and leading difference

$$t - t_{\log} = \frac{(-1)^k\zeta^{(k)}(0)}{2H_k(\tau)^3}\frac{y^2}{T^{3k+3}}\left[1 + O_K\left(\frac{\log T}{T}\right)\right], \tag{6.19}$$

whenever the displayed coefficient is nonzero. If it vanishes, the divisibility statement still holds and the first nonzero term occurs later. A finite inverse-logarithmic truncation cannot resolve (6.19); its remaining logarithmic error is much larger than y^2.

6.4 What is and is not uniform

The theorem allows ε to be positive, negative, zero, or to change sign, provided $\tau = \varepsilon T$ stays in a fixed compact real interval. It gives no uniform bound as $|\tau| \to \infty$, as $k \to \infty$, or as a complex τ approaches a zero of H_k. It also does not state that the first two terms are accurate at every moderately large T. For example, the recorded diagnostic $k = 2, \tau = -2, T = 30$ lies too early for the first correction to improve the leading approximation. The exact chart and its eventual asymptotics are unaffected; an effective numerical radius would be a separate result.

7 An exact Lambert chart for the dilogarithm

At $s = 2$, formula (3.7) gives

$$\Delta_2(t) = t(1 - \operatorname{Log} t) + t^2 R_*(t), \qquad R_*(t) = \frac{1}{4} - \frac{t}{72} + O(t^3). \tag{7.1}$$

The coefficient of t^2 in R_* is zero. For small positive y, define

$$H = -W_{-1}(-y/e), \qquad x = \frac{y}{H} = e^{1-H}. \tag{7.2}$$

Then $x(1 - \log x) = y$, and $H \to \infty$. The real branch W_{-1} is forced by the small solution. The branch convention and the classical large-argument expansions are described in DLMF 4.13 [2].

Proposition 7.1 (A convergent correction to the exact Lambert core)**.** *There is a holomorphic $V(x, h)$ near $(0, 0)$, $V(0, 0) = 1$, such that*

$$t = xV(x, H^{-1}).$$

It is characterized by

$$1 = v(1 - h \log v) + hxv^2 R_*(xv). \tag{7.3}$$

Moreover,

$$t = x - \frac{x^2}{4(H-1)} + \frac{4H^2 + 28H - 23}{288(H-1)^3} x^3 + O(x^4/H). \tag{7.4}$$

Proof. Substitute $t = xv$ in (7.1) and divide by $xH = y$. The result is (7.3). The derivative in v at $(x, h, v) = (0, 0, 1)$ is one. Thus the implicit function theorem gives V. The solution at $h = 0$, or at $x = 0$, is identically one, so $V - 1$ is divisible by hx.

For fixed small h, write $v = 1 + A_1(h)x + A_2(h)x^2 + \cdots$. The coefficients at degrees one and two in (7.3) give

$$A_1(h) = -\frac{h}{4(1-h)}, \qquad A_2(h) = \frac{h}{72(1-h)} + \frac{h^2}{8(1-h)^2} + \frac{h^3}{32(1-h)^3}.$$

Their substitution yields (7.4). The remaining Taylor series starts with hx^3 in V, proving the stated error after multiplication by x. □

Equation (7.4) keeps infinitely many logarithmic terms inside H, then expands in the exponentially smaller x. It is a useful concrete example of why a finite expansion only in $1/\log(1/y)$ misses an entire level of information.

8 Global control for ordinary inverse coefficients

Let $s \in \{2, 3, \dots\}$, $\rho = \zeta(s)$, and let $\ell_s(w) = \sum_{n\geq 1} a_n^{(s)} w^n$ be the compositional inverse at zero. Lagrange inversion gives

$$a_n^{(s)} = \frac{1}{2\pi i n} \int_C \frac{dz}{\operatorname{Li}_s(z)^n}, \tag{8.1}$$

where C is a small positively oriented circle. To pass from a local singular expansion to a coefficient theorem, one must show that the rest of the contour contributes less.

Lemma 8.1 (Zero-freeness and univalence)**.** *For every real $s > 1$, Li_s is univalent on $\Lambda = \mathbb{C} \setminus [1, \infty)$. Its only zero there is zero.*

Proof. The classical integral representation [1] is

$$\operatorname{Li}_s(z) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \frac{z}{e^t - z}\, dt.$$

If $\operatorname{Im} z > 0$, the integrand has positive imaginary part, so $\operatorname{Im} \operatorname{Li}_s(z) > 0$. Also

$$\operatorname{Li}_s'(z) = \frac{1}{\Gamma(s-1)} \int_0^\infty \frac{t^{s-2}}{e^t - z}\, dt,$$

whose imaginary part is positive in the upper half-plane. For distinct z_1, z_2 there, the quotient

$$\frac{\operatorname{Li}_s(z_2) - \operatorname{Li}_s(z_1)}{z_2 - z_1} = \int_0^1 \operatorname{Li}_s'(z_1 + u(z_2 - z_1))\, du$$

has positive imaginary part and is nonzero. Reflection proves injectivity below the real axis. The upper, lower, and real images cannot overlap because their imaginary parts have different signs. On $(-\infty, 1)$, Li_s' is strictly positive. This proves global injectivity and, since $\operatorname{Li}_s(0) = 0$, zero-freeness away from zero. □

Lemma 8.2 (The cut lies strictly outside the endpoint circle)**.** *For integer $s \geq 2$ and every $x > 1$,*

$$|\operatorname{Li}_s(x + i0)| = |\operatorname{Li}_s(x - i0)| > \zeta(s). \tag{8.2}$$

On a compact part of the cut bounded away from 1, the gap is uniform.

Proof. Put $x = e^u$, $u > 0$. The upper-bank imaginary part is

$$\operatorname{Im} \operatorname{Li}_s(e^u + i0) = \frac{\pi u^{s-1}}{(s-1)!}. \tag{8.3}$$

For $s = 1$, this follows from $\operatorname{Li}_1(z) = -\operatorname{Log}(1 - z)$; integration in u proves it successively for higher integers, with zero imaginary boundary value at $u = 0$.

For integer $q \geq 2$, integration by parts in the integral representation gives

$$\operatorname{Re} \operatorname{Li}_q(e^u + i0) = -\frac{1}{\Gamma(q-1)} \int_0^\infty t^{q-2} \log|1 - e^{u-t}|\, dt.$$

On $0 < t < u$, use $-\log(e^{u-t} - 1) > -(u - t)$. On $t > u$, use $-\log(1 - e^{u-t}) > e^{u-t}$. The second integral is at least one, since after $t = u + v$, $(u + v)^{q-2} \geq v^{q-2}$. The first integral is greater than $-u^q/q!$. Hence

$$\operatorname{Re} \operatorname{Li}_q(e^u + i0) > 1 - \frac{u^q}{q!}. \tag{8.4}$$

For $s \geq 3$, let

$$U = \left(\frac{\rho(s-1)!}{\pi}\right)^{1/(s-1)}.$$

For $0 < u \leq U$, equation (8.4) with $q = s-1$ shows $\operatorname{Re}\operatorname{Li}_{s-1}(e^u + i0) > 1 - \rho/\pi > 0$. Thus $\operatorname{Re}\operatorname{Li}_s(e^u + i0)$ increases from ρ up to U. For $u \geq U$, the imaginary part in (8.3) is at least ρ, with strict modulus inequality at U as well. For $s = 2$, use $\operatorname{Re}\operatorname{Li}_1(e^u + i0) = -\log(e^u - 1) > 0$ for $u < \log 2$, and $\rho/\pi = \pi/6 < \log 2$. This proves (8.2). Continuity proves the compact uniform gap. □

Proposition 8.3 (Radius and a dented analytic domain)**.** *The inverse ℓ_s is holomorphic for $|w| < \rho$, and its only singularity on $|w| = \rho$ is $w = \rho$. It is holomorphic in a dented disc*

$$\{\, |w| < \rho + \eta, \quad |\arg(w - \rho)| > \phi \,\}, \qquad \eta > 0, \quad 0 < \phi < \frac{\pi}{2}, \tag{8.5}$$

after reducing η if necessary. Its radius of convergence at zero is exactly ρ.

Proof. For integer s, successive integration of the logarithmic identity for $s = 1$ gives

$$\operatorname{Li}_s(z) = -\frac{(\log|z|)^s}{s!} + O((1 + \log|z|)^{s-1}) \qquad (|z| \to \infty)$$

uniformly on the two banks and within the slit plane. Indeed, integrate $\partial_r \operatorname{Li}_s(re^{i\theta}) = \operatorname{Li}_{s-1}(re^{i\theta})/r$ from a fixed radius greater than one; the initial values there are bounded uniformly, including one-sided bank limits. Induction proves the assertion. Only uniform divergence of the modulus is needed here. If a finite point is on the boundary of $\operatorname{Li}_s(\Lambda)$, a sequence of its preimages cannot escape to infinity by this estimate. An interior limiting preimage would give an interior image point by the open mapping theorem. Thus every finite boundary point comes from $z = 1$ or from a bank of $x > 1$. Lemma 8.2 shows that the image boundary has no point of modulus below ρ, and only $w = \rho$ has modulus ρ. Since the image contains zero, a radial first-exit argument shows that it contains the whole disc $|w| < \rho$. By univalence its inverse is holomorphic there.

Near ρ, put $x = 1 + u$. For $s \geq 3$, the upper-bank image is

$$w - \rho = \zeta(s-1)u + o(u) + i\,O(u^{s-1}),$$

whereas for $s = 2$ it is

$$w - \rho = u(1 - \log u) + i\pi u + O(u^2|\log u|).$$

Both bank curves therefore leave ρ in the positive-real direction. Fix a sufficiently small $\phi > 0$. A short initial segment of each bank lies inside the excluded cone $|\arg(w - \rho)| < \phi$. The remaining bank curves escape to infinity and, by Lemma 8.2, have a positive modulus gap above ρ. Choose η below that gap. The domain in (8.5) is connected, contains zero, and has no point on the boundary of $\operatorname{Li}_s(\Lambda)$. It is therefore contained in that image. This establishes the asserted complex continuation without requiring a prior complex extension of the real Lambert chart. Finally, the nonzero logarithmic term for $s \geq 3$, and the $y/\log(1/y)$ behavior for $s = 2$, show that $w = \rho$ is not removable. Therefore the radius is exactly ρ. □

Remark 8.4 (A fixed, rather than fictitiously uniform, contour gap). Equations (8.1) and (8.2) also permit a direct keyhole proof. On a fixed part $x \geq 1 + \delta$ the modulus gap is positive. On the shrinking part near $x = 1$, one must instead use the local expansion. A positive constant gap cannot hold at moving contour endpoints that themselves approach 1. This separation removes an imprecision in the contour wording of the inspected note [4].

9 All-orders coefficient expansions at higher integer orders

Let $s = m + 1 \geq 3$, $\rho = \zeta(m+1)$, and $a = \zeta(m)$. Write

$$\delta = 1 - \frac{w}{\rho}, \qquad y = \rho\delta.$$

The local inverse is $\ell_{m+1}(w) = e^{-t(y)}$, where t is the inverse of $\Delta_{m+1}(t)$. Theorem 4.1 applies with $d = 1$ and $b = (-1)^m/m!$.

We first record the transfer calculation with its normalization. For fixed complex α,

$$[z^n](1-z)^\alpha = \frac{\Gamma(n-\alpha)}{\Gamma(-\alpha)\Gamma(n+1)}. \tag{9.1}$$

The identity extends analytically through integral α. Differentiating q times in α gives coefficients of $(1-z)^\alpha \log^q(1-z)$. The Gamma quotient has a full expansion in inverse powers of n, locally uniformly in α [3]. Because $1/\Gamma(-\alpha)$ has a simple zero at every nonnegative integer j, one logarithmic degree disappears. In particular,

$$[z^n](1-z)^j \log^q(1-z) = \frac{(-1)^{j+q} q\, j!}{n^{j+1}} (\log n)^{q-1} + O\left(\frac{(1+\log n)^{q-2}}{n^{j+1}} + \frac{(1+\log n)^{q-1}}{n^{j+2}} \right) \tag{9.2}$$

for $q \geq 2$; for $q = 1$, the first error term is absent. This can be read directly from (9.1); the full transfer of a local remainder on a dented disc is the standard singularity-analysis theorem [5].

Theorem 9.1 (Full coefficient expansion and its sharp corner law)**.** *For every integer $m \geq 2$, there are real polynomials Q_N, $N \geq m$, such that, for every fixed $M \geq m$,*

$$a_n^{(m+1)} = \rho^{-n} \left\{ \sum_{N=m}^{M} \frac{Q_N(\log n)}{n^{N+1}} + O_M\left(\frac{(1+\log n)^{\lfloor M/(m-1) \rfloor}}{n^{M+2}} \right) \right\}. \tag{9.3}$$

The degrees satisfy

$$\deg Q_N \leq \left\lfloor \frac{N-1}{m-1} \right\rfloor - 1. \tag{9.4}$$

For $q \geq 1$ and $j = 1 + (m-1)q$,

$$[X^{q-1}] Q_j(X) = (-1)^q \frac{q\, j!}{(m!)^q} \frac{1}{(m-1)q+1} \binom{mq}{q} \frac{\rho^j}{a^{mq+1}}. \tag{9.5}$$

In particular,

$$a_n^{(s)} \sim -\frac{\zeta(s)^{s-1-n}}{\zeta(s-1)^s n^s}, \qquad s \geq 3. \tag{9.6}$$

Proof. The convergent expansion of $t(y)$ has at y^N logarithmic degree at most $\lfloor (N-1)/(m-1) \rfloor$. In e^{-t}, a product of h factors from t at total power N has logarithmic degree at most $\lfloor (N-h)/(m-1) \rfloor$. Thus the same bound holds for e^{-t}, and at the extreme power $N = 1 + (m-1)q$ only the linear term $-t$ can contribute logarithmic degree q.

Substitute $y = \rho(1 - w/\rho)$, so that $\log y = \log\rho + \log(1 - w/\rho)$. The constant $\log\rho$ does not affect highest logarithmic degrees. Proposition 8.3, the remainder bound (4.9), and the full Gamma-quotient expansion in (9.1) justify termwise singularity transfer to any fixed order. Analytic polynomial terms at the singular point have no algebraic coefficient contribution. Transfer reduces each positive logarithmic degree by one. Corrections from lower local powers do not increase that degree. This proves (9.3) and (9.4). The stated remainder is a deliberately safe bound; one can often lower its logarithmic exponent.

At $j = 1+(m-1)q$, the extreme local coefficient in e^{-t} is the negative of (4.10), multiplied by ρ^j. Equation (9.2) multiplies it by $(-1)^{j+q}qj!$. Using $b = (-1)^m/m!$ and $j = 1+(m-1)q$, the resulting sign is $(-1)^q$, and the remaining factors give (9.5). No lower local power has enough logarithmic degree to affect it. At $q = 1$, $j = m$, the formula is $Q_m = -\rho^m/a^{m+1}$, which proves (9.6). □

Example 9.2 (Where the first coefficient logarithm occurs)**.** At $q = 2$, the first new highest logarithmic term occurs at $N = 2m-1$, hence at coefficient order n^{-2m}:

$$[X]Q_{2m-1}(X) = \frac{2m(2m-1)!}{(m!)^2}\frac{\rho^{2m-1}}{a^{2m+1}}. \tag{9.7}$$

For the trilogarithm this is $6\zeta(3)^3/\zeta(2)^5$ multiplying $\rho^{-n}n^{-4}\log n$. For the tetralogarithm it is $20\zeta(4)^5/\zeta(3)^7$ multiplying $\rho^{-n}n^{-6}\log n$. Lower inverse powers of n can contain constant corrections, but cannot contain that first logarithm.

9.1 A finite procedure for every Q_N

The theorem does not leave the lower coefficients unspecified in principle. For any target M, the following finite operations compute all of them.

1. Expand the explicit analytic A and logarithmic P from (4.4)–(4.5) to the required power.
2. Solve (4.11) degree by degree, or substitute $t = \sum_{N\le M} y^N T_N(\log y)$ in the forward equation.
3. Expand e^{-t}, then replace $y = \rho\delta$, $\log y = \log\rho + \log\delta$.
4. Apply derivatives of (9.1), expanded to order n^{-M-1}.

All local coefficients lie in the field generated by rational numbers, the finitely many relevant zeta values, $\log\zeta(m)$, and $\log\zeta(m+1)$; the transfer constants introduce Euler's constant and zeta values through Gamma derivatives. This is an explicit symbolic algorithm, not an assertion of a new finite classical-constant evaluation.

10 An all-orders Lambert expansion for inverse dilogarithm coefficients

Here $\rho = \zeta(2)$, $a_n = a_n^{(2)}$, and $n \to \infty$. Define a different Lambert variable from the local one in (7.2):

$$H_n = -W_{-1}\left(-\frac{\rho}{en}\right), \qquad H_n - \log H_n = \log(n/\rho) + 1. \tag{10.1}$$

Thus $H_n \sim \log n$.

Theorem 10.1 (Reciprocal Gamma coefficients)**.** *For each fixed $N \ge 0$,*

$$a_n = -\frac{\rho^{1-n}}{n^2H_n^2}\left\{\sum_{j=0}^{N}\frac{c_j}{H_n^j} + O_N(H_n^{-N-1})\right\}, \tag{10.2}$$

where

$$\boxed{c_j = -[z^{j+1}]\frac{1}{\Gamma(z-j-1)} = -\frac{(1/\Gamma)^{(j+1)}(-j-1)}{(j+1)!}.} \tag{10.3}$$

In particular,

$$\begin{aligned} c_0 &= 1, \\ c_1 &= 3 - 2\gamma, \\ c_2 &= 6 - 11\gamma + 3\gamma^2 - \frac{\pi^2}{2}. \end{aligned} \tag{10.4}$$

The expansion is asserted to be asymptotic to every fixed order. No convergence of $\sum c_j H^{-j}$ *is asserted.*

Proof. Deform the Lagrange contour (8.1) to a keyhole. Lemma 8.2 makes the fixed part away from $x = 1$ exponentially smaller than ρ^{-n}. On the upper bank, for $u = x - 1 > 0$,

$$\mathrm{Li}_2(1 + u + i0) = \rho + u(1 - \log u) + i\pi u + O(u^2|\log u|). \tag{10.5}$$

The two banks in the keyhole have opposite orientations, so their contribution is

$$\frac{1}{\pi n} \int_1^{1+\eta} \mathrm{Im}\{\mathrm{Li}_2(x + i0)^{-n}\}\, dx, \tag{10.6}$$

up to an exponentially smaller error, for a fixed sufficiently small $\eta > 0$.

Set $H = H_n$ and $u = \rho v/(nH)$. The defining relation for H gives $1 - \log u = H - \log v$. Uniformly on a suitable growing local range,

$$\left(\frac{\mathrm{Li}_2(1 + u + i0)}{\rho}\right)^{-n} = e^{-v} \exp\left(\frac{v \log v - i\pi v}{H}\right) + \text{an integrable error.} \tag{10.7}$$

After the normalization in (10.2), the integrated error is $O(H/n)$, which is smaller than every fixed inverse power of H. Indeed, take $0 < v \le e^{H/4}$. The error in the exponent from the quadratic logarithm expansion is bounded by Cv^2/n times a fixed polynomial in $1 + |\log v|/H$; the integrand retains an e^{-cv} bound. Beyond this range, the monotonicity of $u(1 - \log u)$ for small positive u gives $O(\exp(-ce^{H/4}))$. The remaining fixed contour has the modulus gap already proved. This also justifies replacing the local integration range by the half-line when expanding to a fixed order.

It remains to expand, in the asymptotic sense,

$$\int_0^\infty e^{-v} \exp\left(\frac{v \log v}{H}\right) \frac{\sin(\pi v/H)}{\pi/H}\, dv. \tag{10.8}$$

Expression (10.8) is a device for its local expansion: for a fixed finite H, the exponentially increasing factor need not define a convergent integral on the entire half-line. The preceding cutoff argument, followed by termwise integration, is the meaning intended here.

More explicitly, write $h = 1/H$ and use the entire identity

$$\frac{\sin(\pi v h)}{\pi h} = v \int_0^1 \cos(\pi v h u)\, du.$$

Differentiating $e^{hv \log v}$ times this expression at most $N + 1$ times gives, on $v \le e^{H/4}$, the Taylor remainder H^{-N-1} times an integrable majorant of the form

$$C_N e^{-v/2} v^{N+2} (1 + |\log v|)^{N+1}.$$

Near zero, the powers of v make the logarithms integrable. Integrating the finite Taylor coefficients and extending each of those polynomial-logarithmic integrals to infinity is legitimate. The coefficient is

$$c_j = -\frac{1}{\pi(j+1)!} \mathrm{Im} \left(\frac{d}{da} - i\pi\right)^{j+1} \Gamma(a)\bigg|_{a=j+2}. \tag{10.9}$$

To simplify it, write $r = j + 1$. For real a, $\mathrm{Im}\{e^{-i\pi a}\Gamma(a)\} = -\sin(\pi a)\Gamma(a)$. The Gamma reflection formula [3] gives $\sin(\pi a)\Gamma(a) = \pi/\Gamma(1-a)$. Differentiate r times, and evaluate at $a = r + 1$. The factor $e^{-i\pi(r+1)} = (-1)^{r+1}$, together with the chain-rule factor $(-1)^r$, yields

$$\mathrm{Im}(D_a - i\pi)^r\Gamma(a)\big|_{a=r+1} = \pi(1/\Gamma)^{(r)}(-r).$$

Substitution in (10.9) proves (10.3). The first three values follow by expanding the reciprocal Gamma function at the indicated integers. □

Remark 10.2 (An essential distinction in the proof). The cutoff qualification following (10.8) is necessary. The fixed-H function $e^{-v+v\log v/H}$ eventually grows at infinity. Replacing the asymptotic coefficient procedure by an unqualified improper integral would create a false identity. The convergent object in the local inverse problem is the analytic Lambert chart of Section 7; the coefficient theorem is an outer asymptotic expansion.

10.1 Ordinary logarithms and explicit polynomials

Put $L = \log n$ and $B = \log L + 1 - \log\rho$. The same proof with $u = \rho v/(nL)$ gives

$$a_n = -\frac{\rho^{1-n}}{n^2L^2}\left\{\sum_{j=0}^{N}\frac{P_j(B)}{L^j} + O_N\left(\frac{(1+\log L)^{N+1}}{L^{N+1}}\right)\right\}, \tag{10.10}$$

where the finite polynomial formula is

$$P_j(B) = \sum_{h=0}^{\lfloor j/2\rfloor}\frac{(-1)^{j-h}\pi^{2h}}{(2h+1)!(j-2h)!}\sum_{\ell=0}^{j-2h}\binom{j-2h}{\ell}B^{j-2h-\ell}(-1)^\ell\Gamma^{(\ell)}(j+2). \tag{10.11}$$

Here $P_0 = 1$, and

$$\begin{aligned} P_1(B) &= -2B + 3 - 2\gamma, \\ P_2(B) &= 3(B+\gamma-11/6)^2 - \frac{\pi^2}{2} - \frac{49}{12}. \end{aligned} \tag{10.12}$$

In particular, the first correction in ordinary logarithms is

$$a_n = -\frac{\rho^{1-n}}{n^2(\log n)^2}\left[1 + \frac{-2\log\log n + 1 + 2\log\rho - 2\gamma}{\log n} + O\left(\frac{(\log\log n)^2}{(\log n)^2}\right)\right]. \tag{10.13}$$

The coefficients c_j have an especially simple exact generation rule. Since

$$\frac{1}{\Gamma(z-r)} = \frac{\prod_{\ell=0}^{r}(z-\ell)}{\Gamma(1+z)},$$

and

$$\frac{1}{\Gamma(1+z)} = \exp\left(\gamma z + \sum_{\ell\geq 2}\frac{(-1)^{\ell+1}\zeta(\ell)}{\ell}z^\ell\right),$$

only finitely many coefficients are required to compute any c_j. The verification program uses this finite formula, not numerical differentiation at a Gamma pole.

11 Corrections and the limits of the audit

11.1 The two normalization factors

The version of [4] inspected here has SHA-256

783ed56b45aa2e90cbc26ca92106ccea0ab7
ee577e156ad04308448dc65ddd62.

It defines the inverse by $\mathrm{Li}_s(\ell_s(w)) = w$, with no scaling of the output variable. Thus the following differences are not a matter of competing inverse conventions.

Case	Printed leading coefficient	Corrected leading coefficient
$s = 2$	$-\dfrac{\rho^{-n}}{n^2(\log n)^2}$	$-\dfrac{\rho^{1-n}}{n^2(\log n)^2}$
$s \geq 3$	$-\dfrac{\rho^{s-1-n}}{a^{s-1}n^s}$	$-\dfrac{\rho^{s-1-n}}{a^s n^s}$

Here $\rho = \zeta(s)$, $a = \zeta(s-1)$.

For the dilogarithm, the note's scale $f = f(n)$ satisfies

$$\rho f/n = \log f + 1.$$

The retained expression before the final simplification is

$$-\frac{\rho^{-n}}{nf(\log f + 1)}.$$

Since $f \sim n\log n/\rho$ and $\log f + 1 \sim \log n$, this is $-\rho^{1-n}/(n^2(\log n)^2)$. The factor ρ is lost in the final displayed equivalence on page 4.

For $s \geq 3$, the display immediately after the substitution $z = 1 + \rho t/(an)$, on page 5, contains the Jacobian prefactor

$$\frac{\rho^{1-n}}{2\pi i\, a\, n^2}.$$

The immediately following expansion drops $1/a$. The subsequent leading term inherits that loss. Theorem 9.1 provides an independent derivation with the factor retained.

11.2 The dilogarithmic remainder and the contour gap

The last dilogarithm display on page 4 of [4], immediately before the higher-order case, gives a relative remainder $o(\log\log n/\log n)$. Even after restoring the factor ρ, this is too strong. Equation (10.13) implies

$$\frac{-a_n n^2(\log n)^2\rho^{n-1} - 1}{\log\log n/\log n} \longrightarrow -2.$$

Thus a valid leading-order statement has relative remainder $O(\log\log n/\log n)$, or one may use the explicit higher-order expansion above. The earlier replacement of an estimated big-O term by little-o on the same page also lacks justification.

The fixed positive modulus gap claimed along an outer contour whose endpoints approach 1 cannot be uniform in n: the modulus at those endpoints tends to ρ. Lemma 8.2 and the fixed-keyhole construction in Section 10 supply a repair. These observations concern the identified source version; the corrected coefficient theorems here have independent proofs.

11.3 A local consistency check that does not assume the contour proof

The discrepancies can also be detected before doing any global singularity analysis. For $s = m + 1 \geq 3$, the local inverse has

$$\ell_s(w) = \text{analytic part} + \frac{(-1)^m}{m!a^{m+1}}(\rho - w)^m \log(\rho - w) + \cdots .$$

If its coefficients had an asymptotic $-C\rho^{-n}n^{-m-1}$, differentiating $m + 1$ times and applying the elementary Abelian limit $\sum_{n\geq 0} r^n \sim (1-r)^{-1}$ would force $C = \rho^m/a^{m+1}$. This contradicts the printed $C = \rho^m/a^m$. The analogous dilogarithmic check uses the second derivative and $\sum r^n/(\log n)^2 \sim [(1-r)\log^2(1/(1-r))]^{-1}$, obtained by an elementary split at $n \asymp (1-r)^{-1}$. The local Lambert inverse forces $C = \rho$, rather than $C = 1$. The global arguments above establish the corrected asymptotics themselves; the Abelian check separately explains why the printed constants cannot be reconciled with the local inverse.

11.4 Repository corrections versus additions

No concrete mathematical error was found in the directly relevant Polylogarithms resonance formulas that were inspected. Several Transseries reports explicitly state their nonuniform endpoint limitations; those limitations are not errors. The proposed integration changes are therefore additions and a source warning:

1. Add the uniform weighted order-two inverse theorem and its explicit scope to the endpoint research crosswalk.
2. Add the sharp logarithmic support and Fuss–Catalan laws to the local inversion material.
3. Add the inverse coefficient results to the Polylogarithms literature discussion, citing the identified version of [4] with the normalization correction.
4. Preserve the distinction between finite order differentiation, uniform pole cancellation, and the interchange of inversion with parameter differentiation.

The accompanying integration note proposes insertion locations. It does not alter the repository or assert that every existing chapter has been audited.

12 Reproducible verification

The computational files are companions to the proofs. They serve three purposes: detect sign and normalization mistakes, demonstrate the transition quantitatively, and make the symbolic coefficient algorithms reproducible.

12.1 Exact algebraic checks

The exact checks include Bell-polynomial resonance identities, the reciprocal Gamma formula for c_j, and the weighted logarithmic boundary law. For the last test, the program does not merely compare the stated binomial formula with itself. It reverts generic germs whose lower analytic and logarithmic coefficients are all nonzero, then verifies that these perturbations leave the extreme coefficient unchanged.

The dilogarithmic correction in (7.4) is checked by substitution into the forward equation. Small inverse Taylor coefficients are computed independently through Lagrange's formula. The coefficient $[z^{n-1}](z/\operatorname{Li}_s(z))^n/n$ can be obtained using only finite series arithmetic.

12.2 Numerical checks and their meaning

The numerical suite compares exact pole-cancelled formulas with independent evaluations, solves the positive weighted inverse equations, and records leading and corrected approximation errors. For order derivatives near an integer, direct finite differencing of a special-function implementation can inherit its internal integer-order switching error. We therefore use direct summation of the exponentially convergent defining series at moderate positive t as an independent check.

The suite deliberately retains early-asymptotic cases. At $k = 2, \tau = -2, T = 30$, the first correction is less accurate than the leading approximation; at $T = 60$, its truncated value can still be negative. These are limitations of that finite truncation, not of the unique positive exact inverse. They illustrate why an existence proof for a holomorphic chart must not be presented as a numerical accuracy certificate at an arbitrary input.

For the ordinary inverse coefficients, a stable local-cut calculation and finite Lagrange coefficients check the two normalization factors. The all-orders Lambert expansion can also be slow at modest n; adding a term is not guaranteed to improve an asymptotic truncation. The recorded results state the chosen precision, cutoff, and comparison method.

12.3 Selected recorded diagnostics

Calculation	Coverage	Recorded result
Gamma–Bell resonance jets	16	Exact equality
Generic logarithmic reversion	6 families	Exact degree and corner checks
Lambert inverse coefficients	1 formula	Exact through cubic order
Resonant jets versus defining sums	24	Max. absolute error 1.81×10^{-85}
Critical normal form versus defining sums	9	Max. relative error 1.45×10^{-85}
Positive numerical inverse roots	214	Max. normalized residual 4.02×10^{-86}

The inverse solves comprise 27 selected crossover cases, 183 points in the crossover figure, and 4 Lambert cases. Numerical values use 85 decimal digits. Residuals refer to the computed normal-form equation; the reported errors are floating-point diagnostics, not interval bounds.

		$T = 60$		$T = 120$	
k	τ	Leading	First correction	Leading	First correction
0	-2	$+2.57 \times 10^{-1}$	-2.21×10^{-2}	$+1.34 \times 10^{-1}$	-6.76×10^{-3}
0	0	$+8.63 \times 10^{-2}$	-5.95×10^{-3}	$+4.86 \times 10^{-2}$	-1.95×10^{-3}
0	2	$+4.04 \times 10^{-2}$	-2.73×10^{-3}	$+2.21 \times 10^{-2}$	-8.82×10^{-4}
1	-2	$+7.56 \times 10^{-1}$	-2.08×10^{-1}	$+3.75 \times 10^{-1}$	-5.88×10^{-2}
1	0	$+2.92 \times 10^{-1}$	-4.92×10^{-2}	$+1.64 \times 10^{-1}$	-1.65×10^{-2}
1	2	$+9.7 \times 10^{-2}$	-1.48×10^{-2}	$+5.97 \times 10^{-2}$	-5.44×10^{-3}
2	-2	$+1.8$	-1.04	$+8.02 \times 10^{-1}$	-2.58×10^{-1}
2	0	$+7.42 \times 10^{-1}$	-2.69×10^{-1}	$+3.94 \times 10^{-1}$	-8.32×10^{-2}
2	2	$+2.89 \times 10^{-1}$	-7.79×10^{-2}	$+1.73 \times 10^{-1}$	-2.8×10^{-2}

Entries are signed relative errors $(t_{\text{approximation}}/t_{\text{numerical}}) - 1$. An error below -1 means that the truncated approximation is negative, although the numerical inverse is positive.

s	n	Correctly normalized coefficient
2	96	1.03725476316
2	256	1.0660248342
3	96	0.925449222494
3	256	0.961181485931
4	96	0.969541756271
4	256	0.988847869686

The normalization in the last table is $-a_n n^2 H_n^2 \rho^{n-1}$ for $s = 2$, and $-a_n^{(s)} n^s \zeta(s-1)^s \rho^{n-s+1}$ for $s \geq 3$. Each tends to one. The dilogarithmic ratio need not approach one monotonically. Each displayed finite coefficient was evaluated at two precisions; their maximum relative discrepancy was less than 3.09×10^{-129}. This is a rounding diagnostic, not an error bound for its leading asymptotic approximation.

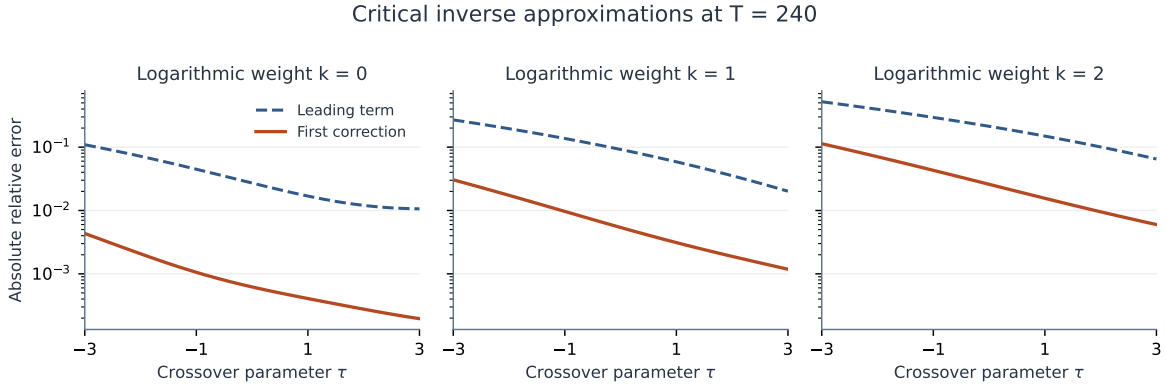


Figure 1: Recorded errors for the weighted critical inverse at $T = 240$. Curves compare finite asymptotic truncations with numerical positive inverses at 183 sampled parameter choices. They illustrate asymptotic accuracy and its dependence on the fixed logarithmic weight; they are not interval-certified bounds.

12.4 Rebuilding the package

The article is built with ordinary pdfLaTeX; its bibliography is embedded. The supplied scripts use Python, mpmath, SymPy, and Matplotlib, with versions listed in `verification/requirements.txt`. The default verification output goes under `build/verification`; it does not overwrite the recorded data. The README gives the exact commands. The recorded checks are finite algebraic equalities or high-precision diagnostics. Neither floating-point residuals nor agreement of two implementations substitutes for the quantified proofs in this article.

13 Further research questions

1. Uniform coefficient extraction at the critical order. Theorem 6.1 is an inverse-germ theorem for εT bounded. Derive an analogue of Theorem 10.1 in a joint limit of the coefficient index and the order, retaining the entire function H_k. The relevant coefficient scale need not be obtained by simply replacing T with $\log n$.

2. The nonlinear feedback endpoint with logarithmic weights. Insert the cancelled normal form into the repository's critical feedback equations and prove the corresponding coefficient and cutoff theorems. The bare inverse established here is one analytic ingredient; the conditioning and global coefficient analysis remain separate tasks.

3. Growing logarithmic weights. Allow $k = k(T)$ with k/T or $k/\log T$ of order one. The positivity of H_k persists on the real axis, but the implicit-function neighborhoods used here are not uniform in k. The integral defining H_k suggests a second saddle analysis.

4. Large-order behavior of the reciprocal Gamma coefficients. Determine the growth, signs, and optimal truncation of

$$c_j = -[z^{j+1}]\Gamma(z - j - 1)^{-1}.$$

The differentiation order and the negative evaluation point grow together. A fixed-point Taylor estimate for $1/\Gamma$ does not settle this problem. Determine whether a suitable summation or acceleration reconstructs the coefficient expansion.

5. Complex zeros of the crossover profile. Near $H_k(\tau_*) = 0$, construct the replacement for (6.5). The leading normalized forward coefficient vanishes and the normalization $1/H_k$ diverges, so a new balance is required. Classify simple and multiple zeros and the resulting local powers and logarithms.

6. Uniform higher-integer resonance with weights. Combine the exact pair (3.2) with the weighted inverse lift while $s \to m + 1$, $m \geq 2$, and $(s - m - 1)\log(1/y)$ remains bounded. The unweighted order-three feedback case has a repository predecessor; the full weighted inverse coefficient transition remains to be worked out.

7. Noninteger logarithmic weights. For real or complex κ, replace $(\log \nu)^k$ by $(\log \nu)^\kappa$, with the first few terms treated separately. Integer order differentiation is no longer a finite exact operation. Determine when a finite holomorphic chart survives and when only an asymptotic or integral representation is available.

8. Mixed polylogarithmic tails and cancellation of the boundary. For finite sums of dilated polylogarithms and order derivatives, classify the cancellations that remove the highest logarithmic polynomial. Determine the replacement support law when $b = 0$ and the first nonzero singular term occurs at a later power.

9. Global geometry beyond integer orders. Lemma 8.1 holds for every real $s > 1$, but the simple cut-modulus argument was used here for integers. Prove a corresponding sharp endpoint-circle theorem for all real orders, or identify the exact range where it fails. This would extend the coefficient conclusions beyond the resonant cases.

10. Effective radii for the convergent charts. Compute explicit polydisc radii and outward-rounded bounds for the implicit equations in Sections 4 and 6. The existing Cauchy estimates prove existence and error shape; they do not give ready-to-use certified numerical constants.

11. Multiple polylogarithms. Investigate endpoint inverses of one-variable multiple polylogarithms whose local expansions have several independent logarithmic blocks. The general polynomial-logarithmic inverse theorem applies once an exact normal form with a nonzero linear term is established, but the depth dependence and possible cancellations require a separate analysis.

12. Formalization and symbolic certification. A realistic first Lean target is the finite support-counting argument (4.12) and the Fuss–Catalan corner extraction. The analytic convergence, holomorphic implicit-function theorem, and singularity transfer are larger formalization tasks. A symbolic implementation should retain the exact resonant pair until a parameter regime is specified, so it cannot silently replace a uniform expression by a nonuniform finite jet.

A Proof dependencies and mathematical status

Result	Essential input	Type of conclusion
Resonant order jets	Classical Jonquière identity, Gamma recurrence, normal convergence	Exact identity on a slit disc
Weighted inverse and corner law	Holomorphic implicit function theorem, support counting, Lagrange inversion	Convergent chart and exact coefficients
Uniform critical inverse	Pole-cancelled weighted identity, nonvanishing H_k, compact parameter set	Convergent uniform chart for fixed k
First exponential block	Analytic divisibility in the third chart variable	Difference from the complete logarithmic block
Global coefficient domain	Positive-imaginary integral, cut-modulus gap, growth at infinity	Univalence, exact radius, dented continuation
Higher inverse coefficients	Convergent local inverse, Gamma quotient, singularity transfer	All fixed orders of an asymptotic expansion
Dilogarithm coefficients	Fixed keyhole gap, local rescaling, integrable Taylor remainders	All fixed orders in the Lambert scale
Computational files	Exact finite algebra and arbitrary-precision evaluations	Reproducibility and diagnostics

The proposed new content lies in the explicit combinations and conclusions described in Section 1. The table is also a list of what would have to be formalized or independently checked to upgrade the package to a machine-verified development.

B A useful finite-series recurrence

For numerical reproduction of individual inverse coefficients, let

$$f(z) = \frac{\mathrm{Li}_s(z)}{z} = \sum_{j\ge 0} f_j z^j, \qquad f_j = (j+1)^{-s}, \qquad f(z)^{-n} = \sum_{j\ge 0} h_j z^j.$$

Then $h_0 = 1$, and differentiation of $fh' = -nf'h$ gives

$$h_k = \frac{1}{k}\sum_{j=1}^{k}((1-n)j-k)f_j h_{k-j}, \qquad a_n^{(s)} = \frac{h_{n-1}}{n}. \tag{B.1}$$

For integer s, the recurrence can be evaluated exactly over $\mathbb{Q}$. For high-precision floating arithmetic, cancellation can be severe; agreement at two working precisions is a diagnostic against that loss, not a rigorous rounding bound.

References

[1] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, Section 25.12, especially equations 25.12.11–25.12.13. https://dlmf.nist.gov/25.12. Accessed 12 October 2026.

[2] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, Section 4.13, Lambert W-function. https://dlmf.nist.gov/4.13. Accessed 12 October 2026.

[3] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, Sections 5.5 and 5.11, Gamma reflection and asymptotic expansions. https://dlmf.nist.gov/5.5, https://dlmf.nist.gov/5.11.

[4] C. Krattenthaler and T. Rivoal, *On the compositional inverse of polylogarithms*, unpublished author-hosted note, dated 10 February 2025, 6 pp. https://rivoal.perso.math.cnrs.fr/articles/inversepolylog.pdf. Inspected version and normalization corrections are specified in Section 11.

[5] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. Chapters VI and VIII, and Appendix A. Author-hosted text: https://algo.inria.fr/flajolet/Publications/book.pdf.

[6] V. Reshetnikov, ProveIt research collection, *Analysis/Polylogarithms*. Inspected manuscript chapters include `01-foundations.tex`, `07-tornheim-evaluation.tex`, `07-shifted-stieltjes-correlations.tex`, `08-periodic-collision-contacts.tex`, and `11-literature.tex`. https://github.com/VladimirReshetnikov/ProveIt/tree/9ec0f0406afe7a5475b03ff65ed91f7e283979b8/Analysis/Polylogarithms.

[7] ProveIt research collection, *Logarithmic Critical Endpoint Lambert Charts*, especially its general logarithmic inverse theorem. https://github.com/VladimirReshetnikov/ProveIt/tree/9ec0f0406afe7a5475b03ff65ed91f7e283979b8/Analysis/Transseries/docs/series-and-transseries/Logarithmic_Critical_Endpoint_Lambert_Charts.

[8] ProveIt research collection, *Confluent Critical Transseries Across the Exponent-Two Boundary*, especially the pole-cancelled normal form and holomorphic inverse chart. https://github.com/VladimirReshetnikov/ProveIt/tree/9ec0f0406afe7a5475b03ff65ed91f7e283979b8/Analysis/Transseries/docs/series-and-transseries/Confluent_Critical_Transseries_Exponent_Two_Boundary.

[9] ProveIt research collection, *Nonlinear Stokes Transport and Logarithmic Inversion*, especially its analytic inverse-transport theorem. https://github.com/VladimirReshetnikov/ProveIt/tree/9ec0f0406afe7a5475b03ff65ed91f7e283979b8/Analysis/Transseries/docs/series-and-transseries/Nonlinear_Stokes_Transport_Logarithmic_Inversion.

[10] ProveIt research collection, *Slowly Varying Action Tails and Critical Transseries*, Section 8 and Research Question 5. https://github.com/VladimirReshetnikov/ProveIt/tree/9ec0f0406afe7a5475b03ff65ed91f7e283979b8/Analysis/Transseries/docs/series-and-transseries/Slowly_Varying_Action_Tails_Critical_Transseries.